

Generation Interconnection System Impact Study Report

For Queue Project Q18-0928

Hilltop Alte 138 kV

February 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0928. The Interconnection Customer proposes a 211.7 MW solar photovoltaic facility interconnecting at the Hilltop Alte 138 kV substation in the New Jersey area. The request was received on January 23, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 267 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,204,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Manolia 1708 to Sta E Toluca 138kV: 132.9% loading (249 MVA) following the outage of the Magnolia 1708 to White Rock-Nike 138kV; DFAX = 28.1%.

The Marion Drive to White Rock 138kV is overloaded to 128.3% of its 538 MVA emergency rating for the loss of the Blanchard to Hilltop Alte 138kV. The project contributes 99.9% DFAX (211.6 MW) to this overload.

Marion Drive to Mjmeuc Generating Sta 1 138kV: 125.5% loading (577 MVA) following the outage of the Hilltop Alte to Blanchard 138kV line; DFAX = 99.9%.

The Mjmeuc Generating Sta 1 to Brown 138kV is overloaded to 123.6% of its 607 MVA emergency rating for the loss of the Hilltop Alte to Blanchard 138kV. The project contributes 99.9% DFAX (211.6 MW) to this overload.

Locust Grove to Sta E Toluca 138kV line: 121.8% loading (255 MVA) following the outage of the Magnolia 1708 to White Rock-Nike Substation 138kV; DFAX = 28.1%.

The Wst Orange to Blanchard Substation 138kV is overloaded to 120.4% of its 667 MVA emergency rating for the loss of the Aerpuerto to Hilltop Alte 138kV. The project contributes 99.9% DFAX (211.6 MW) to this overload.

For the loss of Hilltop Alte Substation to Blanchard 138kV, the Aerpuerto to Brown 138kV loads to 120.3% of its 667 MVA rating. Project DFAX: 99.9%.

Aerpuerto to Hilltop Alte 138kV Line: 119.1% loading (693 MVA) following the outage of the Blanchard to Hilltop Alte 138kV line; DFAX = 99.9%.

The Hilltop Alte to Blanchard 138kV line is overloaded to 119.1% of its 693 MVA emergency rating for the loss of the Hilltop Alte to Aerpuerto Substation 138kV line. The project contributes 99.9% DFAX (211.6 MW) to this overload.

The Mt Poso to Whippany 138kV is overloaded to 115.5% of its 260 MVA emergency rating for the loss of the W Orange to Magnolia 1708 138kV line. The project contributes 36.5% DFAX (77.4 MW) to this overload.

For the loss of Hilltop Alte to Blanchard 138kV, the Veve Calzada 2002 to Laurel Ave 138kV loads to 111.3% of its 357 MVA rating. Project DFAX: 11.9%.

Laurel Ave to White Rock 138kV: 106.7% loading (284 MVA) following the outage of the Blanchard Substation to Hilltop Alte 138kV; DFAX = 50.0%.

For the loss of Hilltop Alte to Blanchard 138kV, the Veve Calzada 2002 to Laurel Ave 138kV loads to 104.1% of its 381 MVA rating. Project DFAX: 11.9%.

For the loss of White Rock-Nike to 15Th St 138kV, the Lyons to 5075 Hosp. Damas 138kV L loads to 101.2% of its 235 MVA rating. Project DFAX: 12.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Magnolia 1708 to Sta E Toluca 138kV line	Rebuild line with higher-capacity conductor	\$7,549,000	21% (\$1,593,000)
White Rock to Marion Drive 138kV	Rebuild line with higher-capacity conductor	\$629,000	26% (\$167,000)
Marion Drive to Mjmeuc Generating Sta 1 138kV	Rebuild line with higher-capacity conductor	\$1,511,000	26% (\$401,000)
Mjmeuc Generating Sta 1 to Brown 138kV	Rebuild line with higher-capacity conductor	\$2,093,000	26% (\$555,000)
Locust Grove to Sta E Toluca 138kV line	Rebuild line with higher-capacity conductor	\$1,790,000	21% (\$378,000)
Blanchard Substation to W Orange Substation 138kV	Rebuild line with higher-capacity conductor	\$1,598,000	26% (\$424,000)
Aerpuerto to Brown 138kV	Rebuild line with higher-capacity conductor	\$1,937,000	26% (\$513,000)
Hilltop Alte to Aerpuerto 138kV	Rebuild line with higher-capacity conductor	\$1,244,000	26% (\$330,000)
Hilltop Alte to Blanchard 138kV line	Rebuild line with higher-capacity conductor	\$1,667,000	26% (\$442,000)
Mt Poso to Whippany 138kV	Rebuild line with higher-capacity conductor	\$4,858,000	19% (\$941,000)
Veve Calzada 2002 to Laurel Ave 138kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,397,000	24% (\$569,000)
White Rock to Laurel Ave Substation 138kV Line 2	Reconductor line	\$1,131,000	26% (\$300,000)
Veve Calzada 2002 to Laurel Ave 138kV	Reconductor line	\$442,000	24% (\$105,000)
Lyons to 5075 Hosp. Damas 138kV	Reconductor line	\$2,308,000	21% (\$486,000)
	Total allocated to project		\$7,204,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.